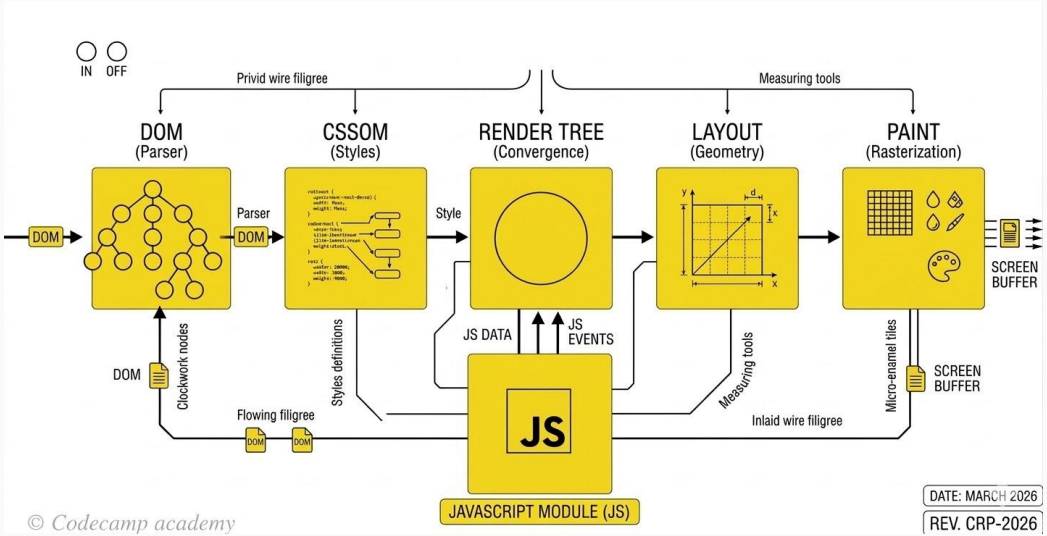
Introduction to how the browsers render Web Pages

Presented by

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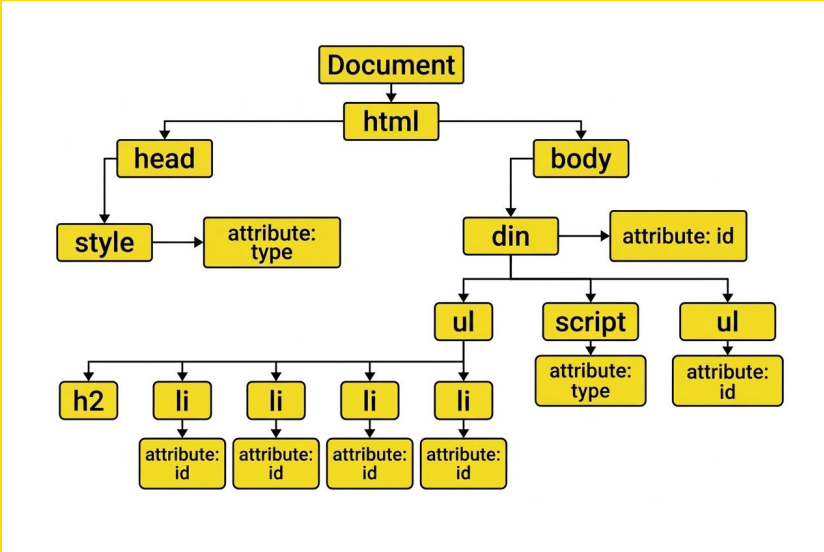
Critical rendering path

When a client sends an HTTP request to a server, the server responds with the resources needed to load the webpage, such as HTML, CSS, and JavaScript files. The browser then processes these resources through a sequence of steps known as the Critical Rendering Path, which determines how the page is parsed, constructed, and finally rendered on the screen.



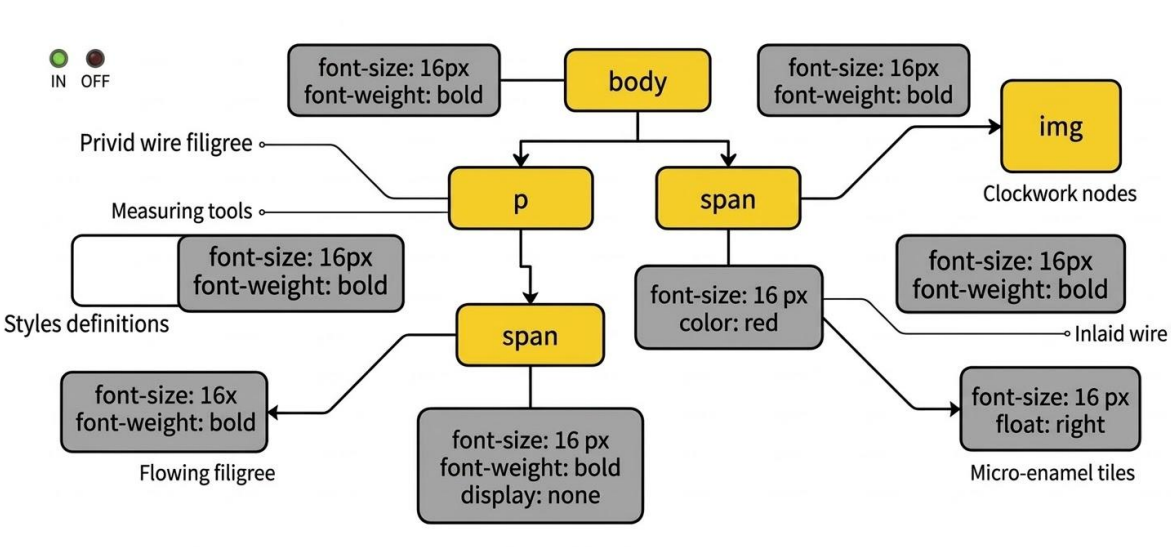
DOM

The DOM (Document Object Model) is created when the browser parses the HTML document. It builds a tree-like structure that represents all the elements, tags, attributes, and content of the webpage. Each element becomes a node in this structure, allowing the browser to understand the hierarchy of the document. The DOM can be accessed and manipulated by JavaScript, enabling dynamic updates to the page without needing to reload it.



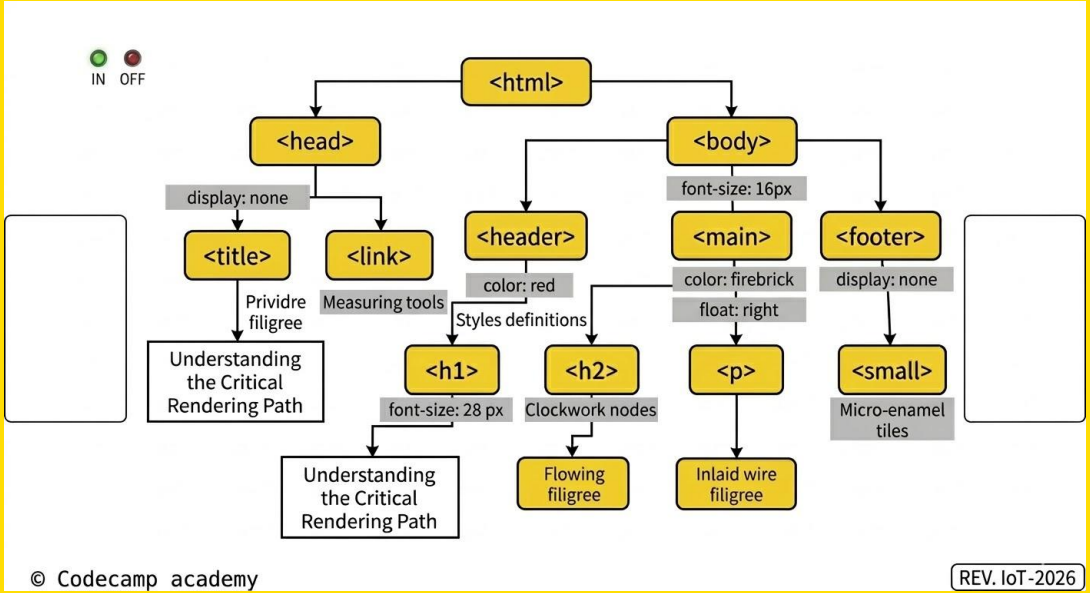
CSSOM

The CSSOM (CSS Object Model) is similar to the DOM but represents the styles of the webpage. The browser parses the CSS files and builds a tree-like structure containing all the style rules and their relationships. This structure maps the styles to the corresponding HTML elements. Once created, the CSSOM combines with the DOM to help the browser determine how each element should be visually displayed on the page.



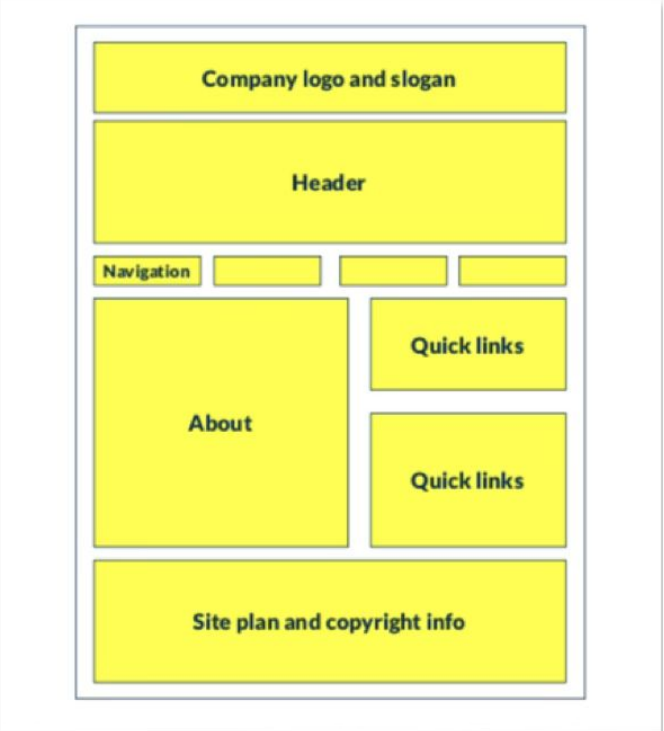
Render tree

The Render Tree is created by combining the DOM and the CSSOM structures. It represents the elements that will actually be displayed on the screen, along with their computed styles. Unlike the DOM, the render tree only includes visible elements. This structure allows the browser to determine how each element should appear before the layout and painting stages.



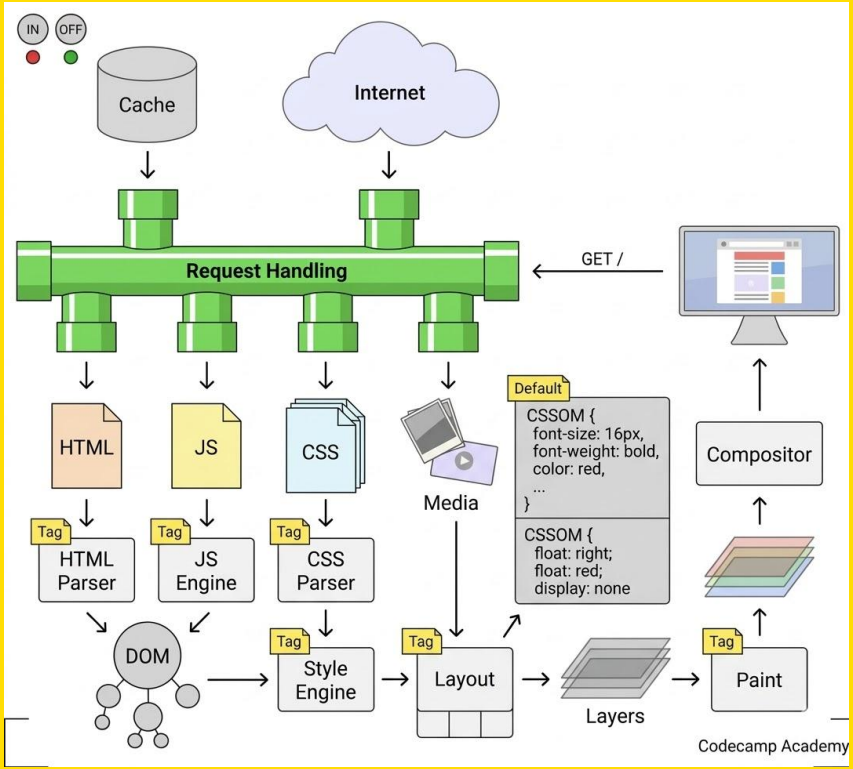
Layout

After the Render Tree is created, the browser performs the Layout step (also known as reflow). In this stage, the browser calculates the size and exact position of each visible element on the page, such as headers, text, images, menus, and footers. It determines how elements are arranged within the viewport based on the computed styles. This step defines the spatial structure of the page before the final visual rendering occurs.

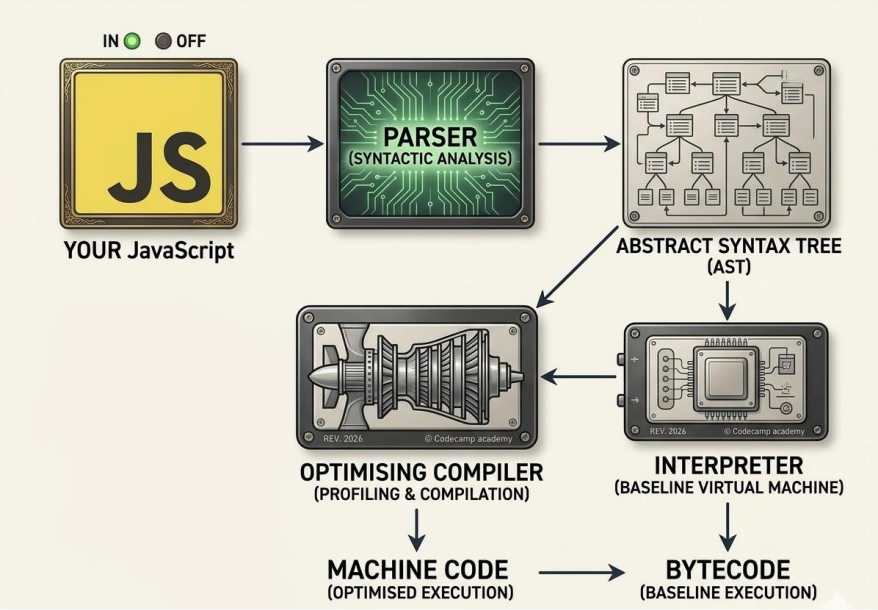


Paint

After the Layout stage is completed, the browser moves to the Paint step. During this phase, the browser fills in the visual details of each element, such as text, colors, images, borders, and shadows. All the information defined in the HTML and CSS is drawn onto the screen based on the calculated layout. This process produces the final visual representation of the webpage for the user.



JavaScript engine



JavaScript runs almost in parallel with the render tree. As a best practice, JavaScript should be placed at the end of the HTML file.

After the browser creates the render tree structure, it executes the JavaScript file at the end.

Because JavaScript is part of web standards, the browser understands the JavaScript language since it has a JavaScript engine developed mainly in C++, among other languages. This engine understands the JavaScript language, which tells the browser what each element is and converts the JavaScript code into machine code.

Thank you!